☰ Fundamentos

En los problemas 1-20, encuentre los extremos relativos de la función indicada.

1. $f(x, y) = x^2 + y^2 + 5$

2. $f(x, y) = 4x^2 + 8y^2$

3. $f(x, y) = -x^2 - y^2 + 8x + 6y$

4. $f(x, y) = 3x^2 + 2y^2 - 6x + 8y$

5. $f(x, y) = 5x^2 + 5y^2 + 20x - 10y + 40$

6. $f(x, y) = -4x^2 - 2y^2 - 8x + 12y + 5$

7. $f(x, y) = 4x^3 + y^3 - 12x - 3y$

8. $f(x, y) = -x^3 + 2y^3 + 27x - 24y + 3$

9. $f(x, y) = 2x^2 + 4y^2 - 2xy - 10x - 2y + 2$

10. $f(x, y) = 5x^2 + 5y^2 + 5xy - 10x - 5y + 18$

11. $f(x, y) = (2x - 5)(y - 4)$

12. $f(x, y) = (x + 5)(2y + 6)$

13. $f(x, y) = -2x^3 - 2y^3 + 6xy + 10$

14. $f(x, y) = x^3 + y^3 - 6xy + 27$

15. $f(x, y) = xy - \dfrac{2}{x} - \dfrac{4}{y} + 8$

16. $f(x, y) = -3x^2y - 3xy^2 + 36xy$

17. $f(x, y) = xe^x \operatorname{sen} y$

18. $f(x, y) = e^{y^2 - 3y + x^2 + 4x}$

19. $f(x, y) = \operatorname{sen} x + \operatorname{sen} y$

20. $f(x, y) = \operatorname{sen} xy$

21. Determine tres números positivos cuya suma sea 21, tal que su producto P sea un máximo. [*Sugerencia*: Exprese P como una función de sólo dos variables.]

22. Determine las dimensiones de una caja rectangular con un volumen de 1 pie^3 que tiene un área superficial mínima S.

23. Encuentre el punto sobre el plano $x + 2y + z = 1$ más cercano al origen. [*Sugerencia*: Considere el cuadrado de la distancia.]

24. Encuentre la distancia mínima entre el punto (2, 3, 1) y el plano $x + y + z = 1$.

25. Encuentre todos los puntos sobre la superficie $xyz = 8$ que son los más cercanos al origen. Determine la distancia mínima.

26. Encuentre la distancia más corta entre las rectas cuyas ecuaciones paramétricas son

$$L_1\colon x = t,\ y = 4 - 2t,\ z = 1 + t,$$

$$L_2\colon x = 3 + 2s,\ y = 6 + 2s,\ z = 8 - 2s.$$

¿En qué puntos sobre las rectas ocurre el mínimo?

27. Determine el volumen máximo de una caja rectangular con lados paralelos a los planos de coordenadas que puede ser inscrito en el elipsoide

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1, \quad a > 0, b > 0, c > 0.$$

28. El volumen de un elipsoide

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1, \quad a > 0, b > 0, c > 0$$

es $V = \frac{4}{3}\pi abc$. Muestre que el elipsoide de mayor volumen que satisface $a + b + c =$ constante es una esfera.

29. El pentágono que se muestra en la FIGURA 13.8.7, formado por un triángulo isósceles sobrepuesto sobre un rectángulo, tiene un perímetro fijo P. Calcule x, y y θ de manera que el área del pentágono sea un máximo.

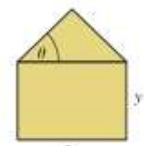


FIGURA 13.8.7 Pentágono del problema 29

30. Un pedazo de latón de 24 pulg de ancho se dobla de manera tal que su sección transversal es un trapezoide isósceles. Vea la FIGURA 13.8.8. Calcule x y θ de manera que el área de la sección transversal sea un máximo. ¿Cuál es el área máxima?

FIGURA 13.8.8 Sección transversal trapezoidal del problema 30

En los problemas 31-34, muestre que la función dada tiene un extremo absoluto pero que el teorema 13.8.2 no es aplicable.

31. $f(x, y) = 16 - x^{2/3} - y^{2/3}$

32. $f(x, y) = 1 - x^4y^2$

33. $f(x, y) = 5x^2 + y^4 - 8$

34. $f(x, y) = \sqrt{x^2 + y^2}$

En los problemas 35-38, encuentre los extremos absolutos de la función continua dada sobre la región cerrada R definida por $x^2 + y^2 \le 1$.

35. $f(x, y) = x + \sqrt{3}y$

36. $f(x, y) = xy$

37. $f(x, y) = x^2 + xy + y^2$

38. $f(x, y) = -x^2 - 3y^2 + 4y + 1$

39. Encuentre los extremos absolutos de $f(x, y) = 4x - 6y$ sobre la región cerrada R definida por $\frac{1}{4}x^2 + y^2 \le 1$.

40. Encuentre los extremos absolutos de $f(x, y) = xy - 2x - y + 6$ sobre la región triangular cerrada R con vértices (0, 0), (0, 8) y (4, 0).

41. La función $f(x, y) = \operatorname{sen} xy$ es continua sobre la región rectangular cerrada R definida por $0 \le x \le \pi$, $0 \le y \le 1$.

a) Encuentre los puntos críticos en la región.

b) Determine los puntos donde f tiene un extremo absoluto.

c) Grafique la función sobre la región rectangular.

Aplicaciones

42. Una función de ingresos es

$$R(x, y) = x(100 - 6x) + y(192 - 4y),$$

donde x y y denotan el número de artículos de dos mercancías vendidas. Dado que la función de costo correspondiente es

$$C(x, y) = 2x^2 + 2y^2 + 4xy - 8x + 20$$

encuentre la ganancia máxima, donde ganancia = ingresos − costo.

43. Se va a construir una caja rectangular cerrada de modo tal que su volumen corresponda a 60 pies^3. El costo del material para la parte superior y el fondo son, respectivamente, de 10 centavos por pie cuadrado y 20 centavos por pie cuadrado. El costo de los lados es de 2 centavos por pie cuadrado. Determine la función de costo $C(x, y)$, donde x y y son la longitud y el ancho de la caja, respectivamente. Calcule las dimensiones de la caja que producirán un costo mínimo.

13.8 Extrema of Multivariable Functions

1. $f_x = 2x$; $f_{xx} = 2$; $f_{xy} = 0$; $f_y = 2y$; $f_{yy} = 2$; $D = 4$. Solving $f_x = 0$ and $f_y = 0$, we obtain the critical point $(0, 0)$. Since $D(0, 0) = 4 > 0$ and $f_{xx}(0, 0) = 2 > 0$, $f(0, 0) = 5$ is a relative minimum.

2. $f_x = 8x$; $f_{xx} = 8$; $f_{xy} = 0$; $f_y = 16y$; $f_{yy} = 16$; $D = 128$. Solving $f_x = 0$ and $f_y = 0$, we obtain the critical point $(0, 0)$. Since $D(0, 0) = 128 > 0$ and $f_{xx}(0, 0) = 8 > 0$, $f(0, 0) = 0$ is a relative minimum.

3. $f_x = -2x + 8$; $f_{xx} = -2$; $fxy = 0$; $f_y = -2y + 6$; $f_{yy} = -2$; $D = 4$. Solving $f_x = 0$ and $f_y = 0$ we obtain the critical point $(4, 3)$. Since $D(4, 3) = 4 > 0$ and $f_{xx}(4, 3) = -2 < 0$, $f(4, 3,) = 25$ is a relative maximum.

4. $f_x = 6x - 6$; $fxx = 6$; $fxy = 0$; $f_y = 4y + 8$; $f_{yy} = 4$; $D = 24$. Solving $f_x = 0$ and $f_y = 0$, we obtain the critical point $(1, -2)$. Since $D(1, -2) = 24 > 0$ and $f_{xx}(1, -2) = 6 > 0$, $f(1, -2) = -11$ is a relative minimum.

5. $f_x = 10x + 20$; $f_{xx} = 10$; $f_{xy} = 0$; $f_y = 10y - 10$; $f_{yy} = 10$; $D = 100$. Solving $f_x = 0$ and $f_y = 0$, we obtain the critical point $(-2, 1)$. Since $D(-2, 1) = 100 > 0$ and $f_{xx}(-2, 1) = 10 > 0$, $f(-2, 1) = 15$ is a relative minimum.

6. $f_x = -8x - 8$; $f_{xx} = -8$; $f_{xy} = 0$; $f_y = -4y + 12$; $f_{yy} = -4$; $D = 32$. Solving $f_x = 0$ and $f_y = 0$, we obtain the critical point $(-1, 3)$. Since $D(-1, 3) = 32 > 0$ and $f_{xx}(-1, 3) = -8 < 0$, $f(-1, 3) = 27$ is a relative maximum.

7. $f_x = 12x^2 - 12$; $f_{xx} = 24x$; $f_{xy} = 0$; $f_y = 3y^2 - 3$; $f_{yy} = 6y$; $D = 144xy$. Solving $f_x = 0$ and $f_y = 0$, we obtain the critical points $(-1, -1)$, $(-1, 1)$, $(1, -1)$, and $(1, 1)$. Since $D(-1, 1) = -144 < 0$ and $D(1, -1) = -144 < 0$, these points do not give relative extrema. Since $D(-1, -1) = 144 > 0$ and $f_{xx}(-1, -1) = -24 < 0$, $f(-1, -1) = 10$ is a

relative maximum. Since $D(1,1) = 144 > 0$ and $f_{xx}(1,1) = 24 > 0$, $f(1,1) = -10$ is a relative minimum.

8. $f_x = -3x^2 + 27$; $f_{xx} = -6x$; $f_{xy} = 0$; $f_y = 6y^2 - 24$; $f_{yy} = 12y$; $D = -72xy$. Solving $f_x = 0$, $f_y = 0$, we obtain the critical points $(-3,-2)$, $(-3,2)$, $(3,-2)$, and $(3,2)$. Since $D(-3,-2) = -432 < 0$ and $D(3,2) = -432 < 0$, these points do no give relative extrema. Since $D(-3,2) = 432 > 0$ and $f_{xx}(-3,2) = 18 > 0$, $f(-3,2) = 432 > 0$ and $f_{xx}(3,-2) = -18 < 0, f(3,-2) = 89$ is a relative maximum.

9. $f_x = 4x - 2y - 10$; $f_{xx} = 4$; $f_{xy} = -2$; $f_y = 8y - 2x - 2$; $f_{yy} = 8$; $D = 32 - (-2)^2 = 28$. Setting $f_x = 0$ and $f_y = 0$, we obtain $4x - 2y = 10$ and $8y - 2x = 2$ or $2x - y = 5$ and $4y - x = 1$. Solving, we obtain the critical point $(3,1)$. Since $D(3,1) = 28 > 0$ and $f_{xx}(3,1) = 4 > 0$, $f(3,1) = -14$ is a relative minimum.

10. $f_x = 10x + 5y - 10$; $f_{xx} = 10$; $f_{xy} = 5$; $f_y = 10y + 5x - 5$; $f_{yy} = 10$; $D = 100 - (5)^2 = 75$. Setting $f_x = 0$ and $f_y = 0$, we obtain $10x + 5y = 10$ and $10y + 5x = 5$ or $2x + y = 2$ and $2y + x = 1$. Solving, we obtain the critical point $(1,0)$. Since $D(1,0) = 75 > 0$ and $f_{xx}(1,0) = 10 > 0$, $f(1,0) = 13$ is a relative minimum.

11. $f_x = 2t - 8$; $f_{xx} = 0$; $f_{xy} = 2$; $f_y = 2x - 5$; $f_{yy} = 0$; $D = 0 - 2^2 = -4$. Since $D(x,y) = -4 < 0$ for all (x,y), there are no relative extrema.

12. $f_x = 2y + 6$; $f_{xx} = 0$; $f_{xy} = 2$; $f_y = 2x + 10$; $f_{yy} = 0$; $D = 0 - 2^2 = -4$. Since $D(x,y) = -4 < 0$ for all (x,y), there are no relative extrema.

13. $f_x = -6x^2 + 6y$; $f_{xx} = -12x$; $f_{xy} = 6$; $f_y = -6y^2 + 6x$; $f_{yy} = -12y$; $D = 144xy - 36$. Setting $f_x = 0$ and $f_y = 0$, we obtain $-6x^2 + 6y = 0$ and $-6y^2 + 6x = 0$ or $y = x^2$ and $x = y^2$. Substituting $x = y^2$ into $y = x^2$, we obtain $y = y^4$ or $y(y^3 - 1) = 0$. Thus, $y = 0$ and $y = 1$. The critical points are $(0,0)$ and $(1,1)$. Since $D(0,0) = -36 < 0$, $(0,0)$ does not give a relative extremum. Since $D(1,1) = 108 > 0$ and $f_{xx}(1,1) = -12 < 0$, $f(1,1) = 12$ is a relative maximum.

14. $f_x = 3x^2 - 6y$; $f_{xx} = 6x$; $f_xy = -6$; $f_y = 3y^2 - 6x$; $f_{yy} = 6y$; $D = 36xy - 36$. Setting $f_x = 0$ and $f_y = 0$, we obtain $3x^2 - 6y = 0$ and $3y^2 - 6x = 0$ or $x^2 = 2y$ and $y^2 = 2x$. Substituting $y = x^2/2$ into $y^2 = 2x$ we obtain $x^4 = 8x$ or $x(x^3 - 8) = 0$. Thus, $x = 0$ and $x = 2$. The critical points $(0,0)$ and $(2,2)$. Since $D(0,0) = -36 < 0$, $f(0,0)$ is not an extremum. Since $D(2,2) = 108 > 0$ and $f_{xx}(2,2) = 12 > 0$, $f(2,2) = 19$ is a relative minimum.

15. $f_x = y + 2/x^2$; $f_{xx} = -4/x^3$; $f_{xy} = 1$; $f_y = x + 4/y^2$; $f_{yy} = -8/y^3; D = 32/x^3Y^3 - 1$. Setting $f_x = 0$ and $f_y = 0$ we obtain $y + 2/x^2 = 0$ and $x + 4/y^2 = 0$. Substituting $y = -2/x^2$ into $x + 4/y^2 = 0$ we obtain $x + x^4 = x(1 + x^3) = 0$. Since $x = 0$ is not in the domain of f, the only critical point is $(-1,-2)$. Since $D(-1,-2) = 3 > 0$ and
$f_{xx}(-1,-2) = 4 > 0$, $f(-1,-2) = 14$ is a relative minimum.

16. $f_x = -6xy - 3y^2 + 36y$; $f_{xx} = -6y$; $f_{xy} = -6x - 6y + 36 = 6(6 - x - y)$; $f_y = -3x^2 - 6xy + 36x$; $f_{yy} = -6x$; $D = 36xy - 36(6 - x - y)^2$. Setting $f_x = 0$ and $f_y = 0$ we obtain $-6xy - 3y^2 + 36y = 0$ and $-3x^2 - 6xy + 36x = 0$ or $-3y(2x + y - 12) = 0$ and $-3x(x + 2y - 12) = 0$. Letting $y = 0$, the first equation is satisfied and the second

equation becomes $-3x(x-12)=0$. Thus, $(0,0)$ and $(12,0)$ are critical points. Similarly, letting $x=0$ we obtain the critical point $(0,12)$. Finally solving $2x+y=12$ and $x+2y=12$ we obtain the critical point $(4,4)$. Since $D(0,0)=-36^2<0$, $D(0,12)=-36^2<0$, and $D(12,0)=-36^2<0$, none of these points give relative extrema. Since $D(4,4)=432>0$ and $f_{xx}(4,4)=-24<0$, $f(4,4)=192$ is a relative maximum.

17. $f_x=(xe^x+e^x)\sin y$; $f_{xx}=(xe^x+2e^x)\sin y$; $f_{xy}=(xe^x+e^x)\cos y$; $f_y=xe^x\cos y$; $f_{yy}=-xe^x\sin y$; $D=-xe^{2x}(x+2)\sin^2 y-e^{2x}(x+1)^2\cos^2 y$. Setting $f_x(x,y)=0$ and $f_y(x,y)=0$ we obtain $(xe^x+e^x)\sin y=0$ and $xe^x\cos y=0$. Since $e^x>0$ for all x, we have $(x+1)\sin y=0$ and $x\cos y=0$. When $x=-1$, we must have $\cos y=0$ or $y=\pi/2+k\pi$, k an integer. When $x=0$, we must have $\sin y=0$ or $y=k\pi$, k an integer. Thus, the critical points are $(0,k\pi)$ and $(-1,\pi/2+k\pi)$, k an integer. Since $D(0,k\pi)=0-\cos^2 k\pi<0$, $(0,k\pi)$ does not give a relative extrema. Now, $D(-1,\pi/2+k\pi)=e^{-2}\sin^2(\pi/2+k\pi)-0>0$ and $f_{xx}(-1,\pi/2+k\pi)=e^{-1}\sin(\pi/2+k\pi)$. Since $f_{xx}(-1,\pi/2+k\pi)$ is positive for k even and negative for k odd, $f(-1,\pi/2+k\pi)=-e^{-1}$ are relative minima for k even, and $f(-1,\pi/2+k\pi)=e^{-1}$ are relative maxima for k odd.

18. $f_x=(2x+4)e^{y^2-3y+x^2+4x}$; $f_{xx}=[(2x+4)^2+2]e^{y^2-3y+x^2+4x}$; $f_{xy}=(2x+4)(2y-3)e^{y^2-3y+x^2+4x}$; $f_y=(2y-3)e^{y^2-3y+x^2+4x}$; $f_{yy}=[(2y-3)^2+2]e^{y^2-3y+x^2+4x}$; $D=[(2x+4)^2+2][(2y-3)^2+2]\cdot e^{2(y^2-3y+x^2+4x)}-[(2x+4)(2y-3)]^2e^{2(y^2-3y+x^2+4x)}$. Setting $f_x=0$ and $f_y=0$ and using the fact that an exponential function is always positive, we obtain $2x+4=0$ and $2y-3=0$. Thus, a critical point is $(-2,3/2)$. Since $D(-2,3/2)=4e^{2(9/4-9/2+4-8)}>0$ and $f_{xx}(-2,3/2)=2e^{9/4-9/2+4-8}>0$, $f(-2,3/2)=e^{9/4-9/2+4-8}=e^{-25/4}$ is a relative minimum.

19. $f_x=\cos x$; $f_{xx}=-\sin x$; $f_{xy}=0$; $f_y=\cos y$; $f_{yy}=-\sin y$; $D=\sin x\sin y$. Solving $f_x=0$ and $f_y=0$, we obtain the critical points $(\pi/2+m\pi,\pi/2+n\pi)$ for m and n integers. For m even and n odd or m odd and n even, $D<0$ and no relative extrema result. For m and n both even, $D>0$ and $f_{xx}<0$ and $f(\pi/2+m\pi,\pi/2+n\pi)=2$ are relative maxima. For m and n both odd, $D>0$ and $fxx>0$ and $f(\pi/2+m\pi,\pi/2+n\pi)=-2$ are relative minima.

20. $f_x=y\cos xy$; $f_{xx}=-y^2\sin xy$; $f_{xy}=-xy\sin xy+\cos xy$; $f_y=x\cos xy$; $f_{yy}=-x^2\sin xy$; $D=x^2y^2\sin^2 xy-(-xy\sin xy+\cos xy)^2=2xy\sin xy\cos xy-\cos^2 xy$. Setting $f_x=0$ and $f_y=0$ we see that $(0,0)$ is a critical point. Also, solving $\cos xy=0$ we obtain $xy=\pi/2+k\pi$ or $y=\pi(1+2k)/2x$ for k an integer. Since $D(0,0)=-1<0$, $(0,0)$ does not give a relative extrema. For any of the critical points $(x,\pi(1+2k)/2x)$, $D=0$ and no conclusion can be drawn from the second partials test. Since $-1\le\sin xy\le 1$ for all $(x,y)f(x,\pi(1+2k)/2x)=-1$ for k odd are relative minima and $f(x,\pi(1+2k)/2x)=1$ for k even are relative maxima.

21. Let the numbers be x,y, and $21-x-y$. We want to maximize $P(x,y)=xy(21-x-y)=21xy-x^2y-xy^2$. Now $P_x=21y-2xy-y^2$; $P_{xx}=-2y$; $P_{xy}=21-2x-2y$; $P_y=21x-x^2-2xy$; $P_{yy}=-2x$; $D=4xy-(21-2x-2y)^2$. Setting $P_x=0$ and $P_y=0$, we obtain $y(21-2x-y)=0$ and $x(21-x-2y)=0$. Letting $x=0$ and $y=0$, we obtain the critical points $(0,0)$, $(0,21)$, and $(21,0)$. Each of these results in $P=0$ which is clearly not a maximum. Solving $21-2x-y=0$ and $21-x-2y=0$, we obtain the critical point $(7,7)$.

Since $D(7,7) = 147 > 0$ and $P_{xx}(7,7) = -14 < 0$, $P(7,7) = 343$ is a maximum. The three numbers are 7,7, and 7.

22. Let the sides of the base of the box by x and y. Then, since the volume of the box is 1, its height is $1/xy$ and $S = 2xy + 2x(1/xy) + 2y(1/xy) = 2xy + 2/y + 2/x$, $x > 0$, $y > 0$. Now $S_x = 2y - 2/x^2$; $S_{xx} = 4/x^3$; $S_{xy} = 2$; $S_y = 2x - 2/y^2$; $S_{yy} = 4/y^3$; $D = 16/x^3y^3 - 4$. Setting $S_x = 0$ and $S_y = 0$ we obtain $y = 1/x^2$ and $x = 1/y^2$. The critical point is $(1,1)$. Since $D(1,1) = 12 > 0$ and $S_{xx}(1,1) = 4 > 0$, $S(1,1) = 6$ is a minimum. The box is 1 foot on each side.

23. Let $(x, y, 1-x-2y)$ be a point on the plane $x + 2y + z = 1$. We want to minimize $f(x,y) = x^2 + y^2 + (1-x-2y)^2$. Now $f_x = 2x - 2(1-x-2y)$; $f_{xx} = 4$; $f_{xy} = 4$; $f_y = 2y - 4(1-x-2y)$; $f_{yy} = 10$; $D = 40 - 4^2 = 24$. Setting $f_x = 0$ and $f_y = 0$ we obtain $2x - 2(1-x-2y) = 0$ and $2y - 4(1-x-2y) = 0$ or $2x + 2y = 1$ and $2x + 5y = 2$. Thus, $(1/6, 1/3)$ is a critical point. Since $D = 24 > 0$ and $f_{xx} = 4 > 0$ for all (x,y), $f(1/6, 1/3) = 1/6$ is a minimum. Thus, the point on the plane closest to the origin is $(1/6, 1/3, 1/6)$.

24. Let $(x, y, 1-x-y)$ be a point on the plane $x + y + z = 1$. We want to minimize the square of the distance between the point and the plane. This is given by

$$f(x,y) = (x-2)^2 + (y-3)^2 + (-x-y)^2 = 2x^2 + 2y^2 - 4x - 6y + 2xy + 13.$$

$f_x = 4x - 4 + 2y$; $f_{xx} = 4$; $f_{xy} = 2$; $f_y = 4y - 6 + 2x$; $f_{yy} = 4$; $D = 16 - 2^2 = 12$. Setting $f_x = 0$ and $f_y = 0$ we obtain $4x - 4 + 2y = 0$ and $4y - 6 + 2x = 0$ or $2x + y = 2$ and $x + 2y = 3$. Thus, $(1/3, 4/3)$ is a critical point. Since $D = 12 > 0$ and $f_{xx} = 4 > 0$ for all (x,y), $f(1/3, 4/3) = 25/3$ is a minimum. Thus, the least distance between the point and the plane is $\sqrt{25/3} = 5/\sqrt{3}$.

25. Let $(x, y, 8/xy)$ be a point on the surface. We want to minimize the square of the distance to the origin or $f(x,y) = x^2 + y^2 + 64/x^2y^2$. Now $f_x = 2x - 128/x^3y^2$; $f_{xx} = 2 + 384/x^4y^2$; $f_{xy} = 256/x^3y^3$; $f_y = 2y - 128/x^2y^3$; $f_{yy} = 2 + 384/x^2y^4$; $D = (2 + 384/x^4y^2)(2 + 384/x^2y^4) - (256)^2/x^6y^6$. Setting $f_x = 0$ and $f_y = 0$ we obtain $2x - 128/x^3y^2 = 0$ and $2y - 128/x^2y^3 = 0$ or $x^4y^2 = 64$ and $x^2y^4 = 64$; $x \neq 0$, $y \neq 0$. This gives $x^4y^2 = x^2y^4$, $x^2y^2(x^2 - y^2) = 0$ or $x^2 = y^2$. Thus, $x^6 = 64$ and $x = \pm 2$. Similarly, $y = \pm 2$ and the critical points are $(-2,-2)$, $(-2,2)$, $(2,-2)$, and $(2,2)$. Since $D(\pm 2, \pm 2) = 48 > 0$ and $f_{xx}(\pm 2, \pm 2) = 8 > 0$, $f(\pm 2, \pm 2) = 12$ are minima. The points closest to the origin are $(-2,-2,2)$, $(-2,2,-2)$, $(2,-2,-2)$, and $(2,2,2)$. The minimum distance is $\sqrt{12} = 2\sqrt{3}$.

26. We will minimize the square of the distance between the lines. This is given by

$$\begin{aligned} f(s,t) &= [(3+2s) - t]^2 + [(6+2s) - (4-2t)]^2 + [(8-2s) - (1+t)]^2 \\ &= (2s - t + 3)^2 + (2s + 2t + 2)^2 + (-2s - t - 7)^2 = 12s^2 + 6t^2 + 8st - 8s + 12t + 62. \end{aligned}$$

$f_s = 24s + 8t - 8$; $f_{ss}24$; $f_{st} = 8$; $f_t = 12t + 8s - 12$; $f_{tt} = 12$; $D = 24(12) - 64 = 224$. Solving $24s + 8t - 8 = 0$ and $12t + 8s - 12 = 0$ we obtain the critical point $(0,1)$. Since $D(0,1) = 224 > 0$ and $f_{ss}(0,1) = 24 > 0$, we see that $f(0,1)$ is a minimum. The corresponding points on the lines are $(3,6,8)$ on $\mathcal{L}_2$ and $(1,2,2)$ on $\mathcal{L}_1$. The minimum distance is $\sqrt{f(0,1)} = \sqrt{56} = 2\sqrt{14}$.

27. We will maximize the square of the volume of the box in the first octant,

$$V(x, y) = x^2Y^2z^2 = x^2y^2(c^2 - c^2x^2/a^2 - c^2y^2/b^2).$$

$V_x = 2c^2xy^2 - 4c^2x^3y^2/a^2 - 2c^2xy^4/b^2$; $V_{xx} = 2c^2y^2 - 12c^2x^2y^2/a^2 - 2c^2y^4/b^2$; $V_{xy} = 4c^2xy - 8c^2x^3y/a^2 - 8c^2xy^3/b^2$; $V_y = 2c^2x^2y - 2c^2x^4/a^2 - 4c^2x^2y^3/b^2$; $V_{yy} = 2c^2x^2 - 2c^2x^4/a^2 - 12c^2x^2y^2/b^2$; $D = V_{xx}V_{yy} - V_{xy}^2$. Setting $V_x = 0$ and $V_y = 0$ we obtain $xy^2 - 2x^3y^2/a^2 - xy^4/b^2 = 0$, $x^2y - x^4y/a^2 - 2x^2y^3/b^2 = 0$, or, assuming $x > 0$ and $y > 0$, $2b^2x^2 + a^2y^2 = a^2b^2$. Solving, we obtain $x^2 = a^2/3$ and $y^2 = b^2/3$. Thus, $(a/\sqrt{3}, b/\sqrt{3})$ is a critical point. Since

$$D(a/\sqrt{3}, b/\sqrt{3}) = (-\frac{14}{9}b^2c^2)(-\frac{14}{9}a^2c^2) - (-\frac{4}{9}abc^2)^2 = \frac{20}{9}a^2b^2c^4 > 0$$

and $v_{xx} = -\frac{14}{9}b^2c^2 < 0$, $V(a/\sqrt{3}, b/\sqrt{3}) = a^2b^2c^2/27$. The maximum volume is

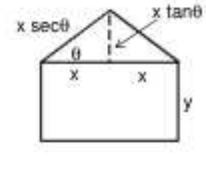
$$8\sqrt{V(a/\sqrt{3}, b/\sqrt{3})} = 8\sqrt{3}abc/9.$$

28. Let $a + b + c = k$. Then $c = k - a - b$ and we want to maximize

$$V(a, b) = 4\pi ab(k - a - b)/3 = 4\pi(kab - a^2b - ab^2)/3.$$

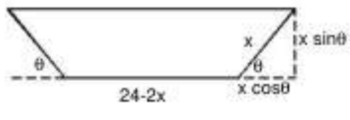
$V_a = \frac{4\pi}{3}(kb - 2ab - b^2)$; $V_{aa} = -\frac{8\pi}{3}$; $V_{ab} = \frac{4\pi}{3}(k - 2a - 2b)$; $V_b = \frac{4\pi}{3}(ka - a^2 - 2ab)$; $V_{bb} = -\frac{8\pi}{3}a$; $D = \frac{64x^2}{9}ab - \frac{16x^2}{9}(k - 2a - 2b)^2$. Setting $V_a = 0$ and $V_b = 0$ we obtain $kb - 2ab - b^2 = 0$ or $ka - a^2 - 2ab = 0$, $a \neq 0$, $b \neq 0$, or $2a + b = k$ and $a + 2b = k$. Solving, we get $a = b = k/3$. Since $D(k/3, k/3) = 16\pi^2k^2/27 > 0$ and $V_{aa}(k/3, k/3) = -8\pi k/9 < 0$, the volume is maximized when $a = b = k/3$. Since $c = k - a - b = k/3$, $a = b = c$ and the ellipsoid is a sphere.

29. The perimeter is given by $P = 2x + 2y + 2x\sec\theta$ and the area is $a = 2xy + x^2\tan\theta$. Solving P for $2y$ and substituting in A, we obtain $A = Px - 2x^2(1 + \sec\theta) + x^2\tan\theta$. Now $A_x(x, \theta) = P - 4x(1 + \sec\theta)2x\tan\theta$; $A_{xx}(x\theta) = -4(1 + \sec\theta) + 2\tan\theta$; $A_{x\theta}(x, \theta) = -4x\sec\theta\tan\theta + 2x\sec^2\theta$; $A_\theta(x\theta) = x^2\sec\theta(\sec\theta - 2\tan\theta)$; $A_{\theta\theta}(x, \theta) = 2x^2\sec\theta(\tan\theta - 2\sec^2\theta + 1)$. We assume that $x > 0$ and $0 \leq \theta \leq \pi/2$.



Setting $A_x = 0$ and $A_\theta = 0$, we obtain $P - 4x(1 + \sec\theta) + 2x\tan\theta = 0$ and $x^2\sec\theta(\sec\theta - 2\tan\theta) = 0$. We note from the second equation and the fact that $\sec\theta \neq 0$ for all θ that $\sec\theta - 2\tan\theta = 0$. Solving for θ, we obtain $\theta = 30°$ and solving $A_x = 0$ for x, we obtain $x_1 = P/(4 + 2\sqrt{3})$. Since $D(x_0, 30°) = (-2\sqrt{3} + 2)(4x_0^2(\sqrt{3} - 5)/3\sqrt{3}) - 0^2 > 9$ and $A_{xx} = 2 - 2\sqrt{3} < 0$, $A(x_0, 30°)$ is a maximum. Letting $x = P/(4 + 2\sqrt{3})$ and $\theta = 30°$ in $P = 2x + 2y + 2x\sec\theta$, we obtain $P = 2y + P?\sqrt{3}$. Thus, the area is maximized for $x = P/(4 + 2\sqrt{3})$, $y = P(\sqrt{3} - 1)/2\sqrt{3}$, and $\theta = 30°$

30. We want to maximize $A(x,\theta) = (x\sin\theta)(24 - 2x + x\cos\theta) = 24x\sin\theta - 2x^2\sin\theta + \frac{1}{2}x^2\sin 2\theta$. Now $A_x = 24\sin\theta - 4x\sin\theta + x\sin 2\theta$; $A_{xx} = -4\sin\theta + \sin 2\theta$; $A_{x\theta} = 24\cos\theta - 4x\cos\theta + 2x\cos 2\theta$; $A_\theta = 24x\cos\theta - 2x^2\cos\theta + x^2\cos 2\theta$; $A_{\theta\theta} = -24x\sin\theta + 2x^2\sin\theta - 2x^2\sin 2\theta$.



We assume $0 < x < 12$ and $=< \theta < \pi/2$. Setting $A_x = 0$ and $A_\theta = 0$ we obtain $24\sin\theta - 4x\sin\theta + 2x\sin\theta\cos\theta = 0$ and $24x\cos\theta - 2x^2\cos\theta + x^2(2\cos^2\theta - 1) = 0$ or $12 - 2x + x\cos\theta = 0$ and $2x\cos^2\theta - 2x\cos\theta + 2\cos\theta - x = 0$. Solving the first equation for $\cos\theta$ and substituting into the second equation, we obtain $2x(2 - 12/x)^2 - 2x(12 - 12/x) + 24(2 - 12/x) - x = 0$. Simplifying, we find $x = 8$ and $\cos\theta = 1/2$ or $\theta = 60°$. Since $D(8, 60°) = (-3\sqrt{3}/2)(-96\sqrt{3}) - (-12)^2 = 288 > 0$ and $A_{xx} = -3\sqrt{3}/2 < 0$, $A(8, 60° = 48\sqrt{3})$ square inches is the maximum area.

31. $f_x = -\frac{2}{3}x^{-1/3}$, $f_y = -\frac{2}{3}y^{-1/3}$. Since $f_x = 0$ and $f_y = 0$ have no solutions, $f(x,y)$ has no critical points and Theorem 13.8.2 does not apply. However, for all (x,y), $f(0,0) = 16 \geq 16 - (x^{1/3})^2 - (y^{1/3})^2 = f(x,y)$, and $f(0,0) = 16$ is an absolute maximum.

32. $f_x = -4x^3y^2$; $f_{xx} = -12x^2y^2$; $f_{xy} = -8x^3y$; $f_y = -2x^4y$; $f_{yy} = -2x^4$; $D = 24x^6y^2 - 64x^6y^2 = -40x^6y^2$. Setting $f_x = 0$ and $f_y = 0$ we see that $(0,y)$ and $(x,0)$ are critical points for any x and y. Since, for any critical point, $D = 0$, Theorem 13.8.2 does not apply. However, for all (x,y), $f(0,0) = 1 \geq 1 - (x^2y)^2 = f(x,y)$, and $f(0,0) = 1$ is an absolute maximum.

33. $f_x = 10x$; $f_{xx} = 10$; $f_{xy} = 0$; $f_y = 4y^3$; $f_{yy} = 12y^2$; $D = 120y^2$. Solving $f_x = 0$ and $f_y = 0$ we obtain the critical point $(0,0)$. Since $D(0,0) = 0$, Theorem 13.8.2 does not apply. However, for any (x,y), $f(0,0) = -8 \leq 5x^2 + y^4 - 8 = f(x,y)$ and $f(0,0) = -8$ is an absolute minimum.

34. $f_x = \frac{x}{\sqrt{x^2+y^2}}$; $f_{xx} = \frac{y^2}{(x^2+y^2)^{3/2}}$; $f_{xy} = -\frac{xy}{(x^2+y^2)^{3/2}}$; $f_y = \frac{y}{\sqrt{x^2+y^2}}$; $f_{yy} = \frac{x^2}{(x^2+y^2)^{3/2}}$; $D = \frac{x^2y^2}{(x^2+y^2)^3} - (\frac{-xy}{(x^2+y^2)^{3/2}})^2 = 0$. Since $D = 0$ for all (x,y), Theorem 13.8.2 does not apply. However, for all (x,y), $f(0,0) = 0 \leq \sqrt{x^2+y^2} = f(x,y)$, so $f(0,0) = 0$ is an absolute minimum.

In Problems 35-38 we parameterize the boundary of R by letting $x = \cos t$ and $y = \sin t$; $0 \leq t \leq 2\pi$. Then, for $(x(t), y(t))$ on the boundary, we maximize or minimize $F(t) = f(\cos t, \sin t)$ on $[0, 2\pi]$.

35. $f_x = 1$; $f_y = \sqrt{3}$. There are no critical points on the interior of R. On the boundary we consider $F(t) = \cos t + \sqrt{3}\sin t$. Solving $F'(t) = -\sin t + \sqrt{3}\cos t = 0$, we obtain critical points at $t = \pi/3$ and $t = 4\pi/3$. Comparing $F(0) = 1$, $F(\pi/3) = 2$, and $F(4\pi/3) = -2$, we see that $f(1/2, \sqrt{3}/2) = 2$ is an absolute maximum and $f(-1/2, -\sqrt{3}/2) = -2$ is an absolute minimum.

36. $f_x = y$; $f_y = x$. Solving $f_x = 0$ and $f_y = 0$ we obtain the critical point $(0, 0)$ with corresponding function value $f(0, 0) = 0$. On the boundary we consider $F(t) = \cos t \sin t = \frac{1}{2} \sin 2t$. Solving $F'(t) = \cos 2t = 0$, we obtain critical points at $\pi/4$, $3\pi/4, 5\pi/4$, and $7\pi/4$. Comparing $f(0, 0,) = 0$, $F(0) = 0$, $F(\pi/4) = 1/2$, $F(3\pi/4) = -1/2$, $F(5\pi/4) = 1/2$, and $F(7\pi/4) = -1/2$, we see that $f(\sqrt{2}/2, \sqrt{2}/2) = f(-\sqrt{2}/2, -\sqrt{2}/2) = 1/2$ are absolute maxima and $f(-\sqrt{2}/2, \sqrt{2}/2) = f(\sqrt{2}/2, -\sqrt{2}/2) = -1/2$ are absolute minima.

37. $f_x = 2x + y$; $f_y = x + 2y$. Solving $f_x = 0$ and $f_y = 0$ we obtain the critical point $(0, 0)$ with corresponding function value $f(0, 0) = 0$. On the boundary we consider $F(t) = \cos^2 t + \cos t \sin t + \sin^2 t = 1 + \frac{1}{2} \sin 2t$. Solving $F'(t) = \cos 2t = 0$ we obtain critical points at $\pi/4$, $3\pi/4$, $5\pi/4$, and $7\pi/4$. Comparing $f(0, 0) = 0$, $F(0) = 1$, $F(\pi/4) = 3/2$; $F(3\pi/4) = 1/2$, $F(5\pi/4) = 3/2$, and $F(7\pi/4) = 1/2$, we see that $f(\sqrt{2}/2, \sqrt{w}/2) = f(-\sqrt{2}/2, -\sqrt{2}/2) = 3/2$ are absolute maxima and $f(0, 0) = 0$ is an absolute minimum.

38. $f_x = -2x$; $f_y = -6y + 4$. Solving $f_x = 0$ and $f_y = 0$, we obtain the critical point $(0, 2/3)$, which is inside R, with corresponding function value $f(0, 2/3) = 5$. On the boundary we consider $F(t) = -\cos^2 t - 3\sin^2 t + 4\sin t + 1$. Solving $F'(t) = 2\cos t \sin t - 6 \sin t \cos t + 4\cos t = 4\cos t - 4\sin t \cos t = 0$, we obtain critical points at $\pi/2$ and $3\pi/2$. Comparing $f(0, 2/3) = 5$, $F(0) = 0$, $F(\pi/2) = 2$, and $F(3\pi/2) = -6$, we see that $f(0, -1) = -6$ is an absolute minimum and $f(0, 2/3) = 5$ is an absolute maximum.

39. $f_x = 4$; $f_y = -6$. There are no critical points over the region R, so absolute extrema must occur on the boundary. We parameterize the boundary by $x = 2\cos t$ and $y = \sin t$ for $0 \le t \le 2\pi$. Considering $F(t) = 8\cos t - 6\sin t$ we obtain $F'(t) = -8\sin t - 6\cos t$. Solving $F'(t) = 0$ we find $\tan t = -3/4$. Using $1 + \tan^2 t = \sec^2 t$ we see that $\sec^2 t = 25/16$ and $\cos t = -4/5$, t is in the second quadrant and $\sin t = 3/5$. The corresponding points on the boundary of R are $(8/5, -3/5)$ and $(-8/5, 3/5)$. Comparing $f(0) = F(2\pi) = f(2, 0) = 8$, $f(8/5, -3/5) = 10$, and $f(-8/5, 3/5) = -10$ we see that the absolute minimum is $f(-8/5, 3/5) = -10$ and the absolute maximum is $f(8/5, -3/5) = 10$.

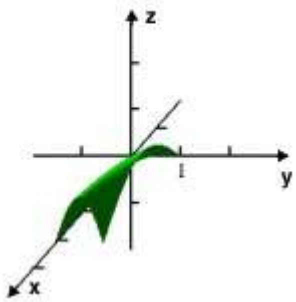
40. $f_x = y - 2$; $f_y = x - 1$. Solving $f_x = 0$ and $f_y = 0$ we obtain the critical point $(1, 2)$ in the region. On $x = 0$, $F(y) = f(0, y) = -y + 6$, which has no critical points for $0 \le y \le 8$. The endpoints of the interval are $(0, 0)$ and $(0, 8)$. On $y = 0$, $G(x) = f(x, 0) = -2x + 6$, which has no critical points for $0 \le x \le 4$. The endpoints of the interval are $(0, 0)$ and $(4, 0)$. On $y = -2x + 8$, $H(x) = f(x, -2x + 8) = x(-2x + 8) = x(-2x + 8) - 2x - (-2x + 8) + 6 = -2x^2 + 8x - 2$. Solving $H'(x) = -4x + 8 = 0$ we obtain $x = 2$. The corresponding point on the triangle is $(2, 4)$. Comparing $f(0, 0) = 6$, $f(0, 8) = -2$; $f(4, 0) = -2$; $f(2, 4) = 6$, and $f(1, 2) = 4$ we see that absolute maxima are $f(0, 0) = f(2, 4) = 6$ and absolute minima are $f(0, 8) = f(4, 0) = -2$.

41. (a) $f_x = y\cos xy$; $f_y = x\cos xy$. Setting $f_x = 0$ and $f_y = 0$ we obtain $y\cos xy = 0$ and $x\cos xy = 0$. If $y = 0$ from the first equation, then necessarily $x = 0$ from the second equation. Thus, $(0, 0)$ is a critical point. For $x \ne 0$ and $y \ne 0$ we have $\cos xy = 0$ or $xy = \pi/2$. Thus, all points $(x, \pi/2x)$ for $0 \le x \le \pi$ are also critical points.

(b) Since $0 \le \sin xy \le 1$ for $0 \le x \le \pi$ and $0 \le y \le 1$, $f(x, y) = \sin xy$ has absolute minima at any points for which $\sin xy = 0$ and absolute maxima at any points for which

$\sin xy = 1$. Thus, $f(x, y)$ has absolute minima when $xy = 0$ or $xy = \pi$, that is, at the points $(0, y)$, $(x, 0)$, and $(\pi, 1)$ which are in the region. Absolute maxima occur when $xy = \pi/2$ or along the curve $y = \pi/2x$ in the region

(c)



42. We want to maximize $P(x, y) = R(x, y) - C(x, y) = 108x - 8x^2 + 192y - 6y^2 - 4xy - 20$. Now $P_x = 108 - 16x - 4y$; $P_{xx} = -16$; $P_{xy} = -4$; $P_y = 192 - 4x$; $P_{yy} = -12$; $D = 192 - 16 = 176$. Setting $P_x = 0$ and $P_y = 0$ we obtain $108 - 16x - 4y = 0$ and $192 - 12y - 4x = 0$ or $4x + y = 27$ and $x + 3y = 48$. Solving, we see that $(3, 15)$ is a critical point. Since $D(3, 15) = 175 > 0$ and $P_{xx}(3, 15) = -16 < 0$, $P(3, 15) = 1582$ is the maximum profit

43. Since the volume of the box is 60, the height is $60/xy$. Then

$$C(x, y) = 10xy + 20xy + 2[2x60/xy + 2y60/xy] = 30xy + 240/y + 240/x.$$

$C_x = 30y - 240/x^2$; $C_{xx} = 480/x^3$, $C_{xy} = 30$; $C_y = 30x - 240/y^2$; $c_{yy} = 480/y^3$; $D = 480^2/x^3y^3 - 900$. Setting $C_x = 0$ and $C_y = 0$ we obtain $30y - 240/x^2 = 0$ and $30x - 240/y^2 = 0$ or $y = 8/x^2$ and $x = 8/y^2$. Substituting the first equation into the second, we have $x = x^4/8$ or $x(x^3 - 8) = 0$. Thus, $(2, 2)$ is a critical point. Since $D(2, 2) = 2700 > 0$ and
$C_{xx}(2, 2) = 60 > 0$, $C(2, 2)$ is a minimum. Thus, the cost is minimized when the base of the box is 2 feet square and the height is 15 feet.

Multiplicadores de lagrange

☰ Fundamentos

En los problemas 1 y 2, dibuje las gráficas de las curvas de nivel de la función f dada y la ecuación de restricción que se indica. Determine si f tiene un extremo con restricciones.

1. $f(x, y) = x + 3y$, sujeta a $x^2 + y^2 = 1$

2. $f(x, y) = xy$, sujeta a $\frac{1}{2}x + y = 1$, $x \geq 0$, $y \geq 0$

En los problemas 3-20, utilice el método de los multiplicadores de Lagrange para encontrar los extremos con restricciones de la función dada.

3. Problema 1

4. Problema 2

5. $f(x, y) = xy$, sujeta a $x^2 + y^2 = 2$

6. $f(x, y) = x^2 + y^2$, sujeta a $2x + y = 5$

7. $f(x, y) = 3x^2 + 3y^2 + 5$, sujeta a $x - y = 1$

8. $f(x, y) = 4x^2 + 2y^2 + 10$, sujeta a $4x^2 + y^2 = 4$

9. $f(x, y) = x^2 + y^2$, sujeta a $x^4 + y^4 = 1$

10. $f(x, y) = 8x^2 - 8xy + 2y^2$, sujeta a $x^2 + y^2 = 10$

11. $f(x, y) = x^3y$, sujeta a $\sqrt{x} + \sqrt{y} = 1$

12. $f(x, y) = xy^2$, sujeta a $x^2 + y^2 = 27$

13. $f(x, y, z) = x + 2y + z$, sujeta a $x^2 + y^2 + z^2 = 30$

14. $f(x, y, z) = x^2 + y^2 + z^2$, sujeta a $x + 2y + 3z = 4$

15. $f(x, y, z) = xyz$, sujeta a $x^2 + \frac{1}{4}y^2 + \frac{1}{9}z^2 = 1$, $x > 0, y > 0, z > 0$

16. $f(x, y, z) = xyz + 5$, sujeta a $x^3 + y^3 + z^3 = 24$

17. $f(x, y, z) = x^3 + y^3 + z^3$, sujeta a $x + y + z = 1$, $x > 0, y > 0, z > 0$

18. $f(x, y, z) = 4x^2y^2z^2$, sujeta a $x^2 + y^2 + z^2 = 9$, $x > 0, y > 0, z > 0$

19. $f(x, y, z) = x^2 + y^2 + z^2$, sujeta a $2x + y + z = 1$, $-x + 2y - 3z = 4$

20. $f(x, y, z) = x^2 + y^2 + z^2$, sujeta a $4x + z = 7$, $z^2 = x^2 + y^2$

21. Encuentre el área máxima de un triángulo rectángulo cuyo perímetro es 4.

22. Encuentre las dimensiones de una caja rectangular abierta con volumen máximo si su área superficial es igual a 75 cm^3. ¿Cuáles son las dimensiones si la caja es cerrada?

☰ Aplicaciones

23. A un tanque cilíndrico recto se le superpone una tapa cónica en la forma que se ilustra en la FIGURA 13.10.6. El radio del tanque es de 3 m y su área superficial total corresponde a 81 πm^2. Encuentre las alturas x y y de manera que el volumen del tanque sea un máximo. [*Sugerencia*: El área superficial del cono es $3\pi\sqrt{9 + y^2}$.]

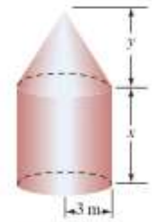


FIGURA 13.10.6 Cilindro con tapa cónica del problema 23

24. En negocios, un índice de utilidad U es una función que produce una medida de la satisfacción obtenida a partir de la compra de cantidades variables, x y y, de dos productos que se venden regularmente. Si $U(x, y) = x^{1/3}y^{2/3}$ es un índice de utilidad, encuentre sus extremos sujetos a $x + 6y = 18$.

25. El **proceso de Haber-Bosch*** produce amoniaco mediante una unión directa de nitrógeno e hidrógeno bajo condiciones de presión P y temperatura constantes:

$$N_2 + 3H_2 \xrightleftharpoons{\text{catalizador}} 2NH_3.$$

Las presiones parciales x, y y z del hidrógeno, nitrógeno y amoniaco satisfacen $x + y + z = P$ y la ley de equilibrio $z^2/xy^3 = k$, donde k es una constante. La cantidad máxima de amoniaco ocurre cuando se obtiene la presión parcial máxima de este mismo. Determine el valor máximo de z.

26. Si una especie de animales tiene n fuentes de alimento, el **índice de amplitud** de su nicho ecológico se define como

$$\frac{1}{x_1^2 + \cdots + x_n^2},$$

donde x_i, $i = 1, 2, \ldots, n$, es la fracción de la dieta de los animales que proviene de la i-ésima fuente de alimentos. Por ejemplo, si la dieta de los pájaros consiste en 50% de insectos, 30% de gusanos y 20% de semillas, el índice de amplitud es

$$\frac{1}{(0.50)^2 + (0.30)^2 + (0.20)^2} = \frac{1}{0.25 + 0.09 + 0.04} = \frac{1}{0.38} \approx 2.63.$$

Advierta que $x_1 + x_2 + \cdots + x_n = 1$ y $0 \leq x_i \leq 1$ para toda i.

***a*)** Para especies con tres fuentes alimenticias, demuestre que el índice de amplitud se maximiza si $x_1 = x_2 = x_3 = \frac{1}{3}$.

***b*)** Demuestre que el índice de amplitud con n fuentes se maximiza cuando $x_1 = x_2 = \cdots = x_n = 1/n$.

***Fritz Haber** (1868-1934) fue un químico alemán. Por el invento de este proceso, Haber obtuvo el premio Nobel de Química en 1918. Carl Bosch, cuñado de Haber e ingeniero químico, fue quien hizo que este proceso fuera práctico a gran escala. Bosch obtuvo el premio Nobel de Química en 1931. Durante la Primera Guerra Mundial el gobierno alemán utilizó el proceso de Haber-Bosch para producir grandes cantidades de fertilizantes y explosivos. Haber fue posteriormente expulsado de Alemania por Adolfo Hitler y murió en el exilio.

Piense en ello

27. Dé una interpretación geométrica de los extremos en el problema 9.

28. Dé una interpretación geométrica de los extremos en el problema 14.

29. Dé una interpretación geométrica del extremo en el problema 19.

30. Dé una interpretación geométrica del extremo en el problema 20.

31. Encuentre el punto $P(x, y)$, $x > 0$, $y > 0$, sobre la superficie $xy^2 = 1$ que es más cercano al origen. Muestre que el segmento de recta del origen a P es perpendicular a la recta tangente en P.

32. Encuentre el valor máximo de $f(x, y, z) = \sqrt[3]{xyz}$ sobre el plano $x + y + z = k$.

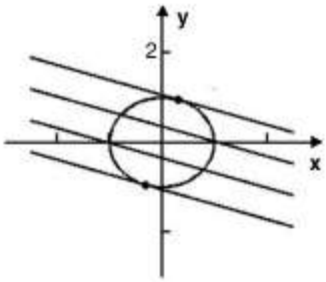
33. Utilice el resultado del problema 32 para probar la desigualdad

$$\sqrt[3]{xyz} \leq \frac{x + y + z}{3}.$$

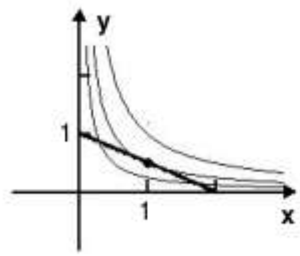
34. Encuentre el punto sobre la curva C de intersección del cilindro $x^2 + z^2 = 1$ y el plano $x + y + 2z = 4$ que está más alejado del plano xz. Encuentre el punto sobre C que es más cercano al plano xz.

13.10 Lagrange Multipliers

1. f has constrained extrema where the level lines intersect the circle.



2. f has constrained extrema where the level curve intersect the line.



3. $f_x = 1$; $f_y = 3$; $g_x = 2x$; $g_y = 2y$. We need to solve $1 = 2\lambda x$, $3 = 2\lambda y$, $x^2 + y^2 - 1 = 0$. Dividing the second equation by the first, we obtain $3 = y/x$ or $y = 3x$. Substituting into the third equation, we have $x^2 + 9x^2 = 1$ or $x = \pm 1/\sqrt{10}$. For $x = 1/\sqrt{10}$, $y = 3/\sqrt{10}$ and for $x = -1/\sqrt{10}$, $y = -3/\sqrt{10}$. A constrained maximum is $f(1/\sqrt{10}, 3/\sqrt{10}) + \sqrt{10}$ and a constrained minimum is $f(-1/\sqrt{10}, -3/\sqrt{10}) = -\sqrt{10}$.

4. $f_x = y$; $f_y = x$; $g_x = 1/2$; $g_y = 1$. We need to solve $y = \lambda/2$, $x = \lambda$, $x/2 + y - 1 = 0$. From the first two equations $y = x/2$. Substituting into the second equation, we have $x = 1$. Thus, $f(1, 1/2) = 1/2$ is a constrained extremum. Since $(0, 1)$ satisfies the constraint and $f(0,1) = 0 < 1/2$, $f(1, 1/2) = 1/2$ is a constrained maximum.

5. $f_x = y$; $f_y = x$; $g_x = 2x$; $g_y = 2y$. We need to solve $y = 2\lambda x$, $x = 2\lambda y$, $x^2 + y^2 - 2 = 0$. Substituting the second equation into the first, we obtain $y = 4\lambda^2 y$ or $y(4\lambda^2 - 1) = 0$. If $y = 0$, then from the second equation $x = 0$. Since $g(0,0) = -2 \neq 0$, $(0,0)$ does not satisfy the constraint. Thus, $\lambda = \pm 1/2$ and $y = \pm x$. Substituting into the third equation, we have $2x^2 = 2$ or $x = \pm 1$. Solutions of the system are $x = 1$, $y = 1$, $\lambda = 1/2$, $x = -1$, $y = -1$; $\lambda = 1/2$, $x = 1$, $y = -1$, $\lambda = -1/2$, and $x = -1$, $y = 1$, $\lambda = -1/2$. Thus, $f(1,1) = f(-1,-1) = 1$ are constrained maxima and $f(1,-1) = f(-1,1) = -1$ are constrained minima.

6. $f_x = 2x$; $f_y = 2y$; $g_x = 2$; $g_y = 1$. We need to solve $2x = 2\lambda$, $2y = \lambda$, $2x + y - 5 = 0$. Substituting the second equation into the first, we find $x = 2y$. Substituting into the third equation, we have $4y + y - 5 = 0$ or $y = 1$. A constrained extremum is $f(2,1) = 5$. Since $(0,5)$ satisfies the constraint and $f(0,5) = 25 > 5$, $f(2,1) = 5$ is a constrained minimum.

7. $f_x = 6x$; $f_y = 6y$; $g_x = 1$; $g_y = -1$. We need to solve $6x = \lambda$, $6y = -\lambda$; $x - y - 1 = 0$. From the first two equations, we obtain $x + y = 0$. Solving this with the third equation, we

obtain $x = 1/2$, $y = -1/2$. Thus, $f(1/2, -1/2) = 13/2$ is a constrained extremum. Since $(1, 0)$ satisfies the constraint and $f(1, 0) = 8 > 13/2$, $f(1/2 - 1/2) = 13/2$ is a constrained minimum.

8. $f_x = 8x$; $f_y = 4y$; $g_x = 8x$; $g_y = 2y$. We need to solve $8x = 8\lambda x$, $4y = 2\lambda y$, $4x^2 + y^2 - 4 = 0$ or $x(\lambda - 1) = 0$, $y(\lambda - 2) = 0$, $4x^2 + y^2 = 4$. If $x = 0$, then from the third equation $y = \pm 2$. If $y = 0$, then from the third equation $x = \pm 1$. The cases $\lambda = 1$ and $\lambda = 2$ lead also to $y = 0$ and $x = 0$, respectively. Thus, $f(0, 2) = f(0, -2) = 18$ are constrained maxima and $f(1, 0) = f(-1, 0) = 14$ are constrained minima.

9. $f_x = 2x$; $f_y = 2y$; $g_x = 4x^3$; $g_y = 4y^3$. We need to solve $2x = 4\lambda x^3$, $2y = 4\lambda y^3$, $x^4 + y^4 - 1 = 0$ or $x(2\lambda x^2 - 1) = 0$, $y(2\lambda y^2 - 1) = 0$, $x^4 + y^4 = 1$. If $x = 0$, the from the third equation $y = \pm 1$. If $y = 0$, then $x = \pm 1$. From $2\lambda x^2 = 1 = 2\lambda y^2$ we have $x^2 = y^2$. Substituting into the third equation, we obtain $x = \pm 1/\sqrt[4]{2}$ and $y = \pm 1/\sqrt[4]{2}$. Solutions of the system are $(0, \pm 1)$, $(\pm 1, 0)$, and $(\pm 1/\sqrt[4]{2}, \pm 1/[4]2)$. Thus, $f(0, \pm 1) = f(\pm 1, 0) = 1$ are constrained minima and $f(\pm 1/\sqrt[4]{2}, \pm 1/\sqrt[4]{2}) = \sqrt{2}$ are constrained maxima.

10. $f_x = 16x - 8y$; $f_y = 4y - 8x$; $g_x = 2x$; $g_y = 2y$. We need to solve $16x - 8y = 2\lambda x$, $4y - 8x = 2\lambda y, x^2 + y^2 - 10 = 0$ or $8 - 47/x = \lambda$, $x^2 + y^2 = 10$. From the first two equations, we obtain $6 - 4y/x = -4x/y$, $6(y/x) - 4(y/x)^2 = -4$, and $2(y/x)^2 - 3(y/x) - 2 = 0$. Factoring, we have $(2y/x + 1)(y/x - 2) = 0$. Then $y = -x/2$ and $y = 2xs$. Substituting $y = -x/2$ into the third equation, we have $x^2 + x^2/4 = 10$ and $x = \pm 2\sqrt{2}$. Substituting $y = 2x$ into the third equation, we have $x^2 + 4x^2 = 10$ and $x = \pm\sqrt{2}$. Solutions of the system are $(2\sqrt{2}, -\sqrt{2})$, $(-2\sqrt{2}, \sqrt{2}), (\sqrt{2}, 2\sqrt{2})$, and $(-\sqrt{2}, -2\sqrt{2})$. Thus, $f(2\sqrt{2}, -\sqrt{2}) = f(-2\sqrt{2}, \sqrt{2}) = 100$ are constrained maxima and $f(\sqrt{2}, 2\sqrt{2}) = f(-\sqrt{2}, -2\sqrt{2}) = 0$ are constrained minima.

11. $f_x = 3x^2y$; $f_y = x^3$; $g_x = 1/2\sqrt{x}$; $g_y = 1/2\sqrt{y}$. We need to solve $3x^2y = \lambda/2\sqrt{x}$, $x^3 = \lambda/2\sqrt{y}$, $\sqrt{x} + \sqrt{y} - 1 = 0$ or $6x^{5/2}y = \lambda$, $2x^3y^{1/2} = \lambda$, $\sqrt{x} + \sqrt{y} = 1$. From the first two equations, we obtain $3x^{5/2}y = x^3y^{1/2}$ and $3\sqrt{y} = \sqrt{x}$. Substituting into the third equation, we have $3\sqrt{y} + \sqrt{y} = 4\sqrt{y} = 1$. Then, $y = 1/16$ and $x = 9/16$. Since $(1/4, 1/4)$ satisfies the constraint and $f(1/4, 1/4) = 1/256$, $f(9/16, 1/16) + 729/65,536$ is a constrained maximum. We also consider $x = 0$, which requires $y = 1$; and $y = 0$, which requires $x = 1$. Since $x \geq 0$ and $y \geq 0$, $f(0, 1) = f(1, 0) = 0 \leq x^3y = f(x, y)$ for all (x, y) which satisfy $\sqrt{x} + \sqrt{y} = 1$. Thus, $f(0, 1) = 0$ and $f(1, 0) = 0$ are constrained minima.

12. $f_x = y^2$; $f_y = 2xy$; $g_x = 2x$; $g_y = 2y$. We need to solve $y^2 = 2\lambda x$, $2xy = 2\lambda y$, $x^2 + y^2 - 27 = 0$ or $y^2 = 2\lambda x$, $y(x - \lambda) = 0$, $x^2 + y^2 = 27$. When $y = 0$ in the third equation, we obtain $x^2 = 27$ or $x = \pm 3\sqrt{3}$, and $\lambda = 0$. When $x = \lambda$ in the second equation, we obtain $y^2 = 2x^2$ from the first equation and $x^2 + 2x^2 = 3x^2 = 27$ from the third equation. This gives $x = \pm 3$ and $y = \pm 3\sqrt{2}$. Since $f(\pm 3\sqrt{3}, 0) = 0$, we see that $f(-3, 3\sqrt{2}) = f(-3, -3\sqrt{2}) = -54$ are constrained minima and $f(3, 3\sqrt{2}) = f(3, -3\sqrt{2}) = 54$ are constrained maxima.

13. $F_x = 1$; $F_y = 2$; $F_z = 1$; $g_x = 2x$; $g_y = 2y$; $g_z = 2z$. We need to solve $1 = 2\lambda x$, $2 = 2\lambda y$; $1 = 2\lambda z$, $x^2 + y^2 + z^2 - 30 = 0$. From the first and second equations, we obtain $y = 2x$. From the first and third equations, we obtain $z = x$. Substituting into the fourth equation, we have $x^2 + 4x^2 + x^2 = 6x^2 = 30$. Thus, $x = \pm\sqrt{5}$, $y = \pm 2\sqrt{5}$, $z = \pm\sqrt{5}$. Then, $F(\sqrt{5}, 2\sqrt{5}, \sqrt{5}) = 6\sqrt{5}$ is a constrained maximum and $F(-\sqrt{5}, -2\sqrt{5}, -\sqrt{5}) = -6\sqrt{5}$ is a constrained minimum.

14. $F_x = 2x$; $F_y = 2y$; $F_x = 2z$; $g_x = 1$; $f_y = 2$; $g_z = 3$. We need to solve $2x = \lambda$, $2y = 2\lambda$, $2z = 3\lambda, x + 2y + 3z - 4 = 0$. From the first and second equations, $y = 2x$. From the first and third equations, $z = 3x$. Substituting into the fourth equation, we have $x+4x+9x = 14x = 4$. Thus, $x = 2/7$, $y = 4/7$, $andz = 6/7$. Then, $F(2/7, 4/7, 6/7) = 56/49$ is a constrained extremum. Since $(4, 0, 0)$ satisfies the constraint and $F(4, 0, 0) = 16 > 56/49$, $F(2/7, 4/7, 6/7) = 56/49$ is a constrained minimum.

15. $F_x = yz$; $F_y = xz$; $F_z = xy$; $g_x = 2x$; $g_y = y/2$; $g_z = 2z/9$. We need to solve $yz = 2\lambda x$, $xz = \lambda y/2$, $xy = 2\lambda z/9$ or $xyz/2 = \lambda x^2$, $xyz/2 = \lambda y^2/4$, $xyz/2 = \lambda z^2/9$ along with $x^2 + y^2/4 + z^2/9 - 1 = 0$ or $x^2 + Y^2/4 + z^2/9 = 1$ for $x > 0$, $y > 0$, $z > 0$. From the first three equations and the fact that $\lambda \neq 0$, $x^2 = y^2/4 = z^2/9$. Substituting into the third equation, we obtain $x^2 + x^2 + x^2 = 3x^2 = 1$, so $x^2 = 1/3$, $y^2 = 4/3$, and $z^2 = 3$. Thus, $F(\sqrt{3}/3, 2\sqrt{3}/3, \sqrt{3}) = 2\sqrt{3}/3$ is a constrained extremum. Since $(\sqrt{23}/6, 1, 1)$ satisfies the constraint and $F(\sqrt{23}/6, 1, 1) = \sqrt{23}/6 < 2\sqrt{3}/3$, $F(\sqrt{3}/3, 2\sqrt{3}/3, \sqrt{3}) = 2\sqrt{3}/3$ is a constrained maximum.

16. $F_x = yz$; $F_y = xz$; $F_z = xy$; $g_x = 3x^2$; $g_y = 3y^2$; $g_z = 3z^2$. We need to solve $yz + 3\lambda x^2$, $xz = 3\lambda y^2$, $xy = 3\lambda z^2$ or $xyz = 3\lambda x^3$, $syz = 3\lambda y^3$, $xyz = 3\lambda z^3$ along with $x^3 + y^3 + z^3 - 24 = 0$ or $x^3 + y^3 + z^3 = 24$. Taking $\lambda = 0$ we see that $(\sqrt[3]{24}, 0, 0)$, $(0, \sqrt[3]{24}, 0)$, and $(0, 0, \sqrt[3]{24})$ satisfy the system. If $\lambda \neq 0$, the the first three equations imply $x^3 = y^3 = z^3$. Substituting into the fourth equation, we obtain $x^3+x^3+x^3 = 3x^3 = 24$ or $x = 2$. Then $(2, 2, 2)$ satisfies the system. Since $\sqrt[3]{24} = 2\sqrt[3]{3}$, $F(2\sqrt[3]{3}, 0, 0) = F(0, 2\sqrt[3]{3}, 0) = F(0, 0, 2\sqrt[3]{3}) = 5$ is a constrained minimum and $F(2, 2, 2) = 13$ is a constrained maximum.

17. $F_x = 3x^2$; $F_y = 3y^2$; $F_z = 3z^2$; $g_x = 1$; $g_y = 1$; $g_z = 1$. We need to solve $3x^2 = \lambda$, $3y^2 = \lambda$, $3z^2 = \lambda$, $x + y + z - 1 = 0$ for $x > 0$, $y > 0$, $z > 0$, and hence $\lambda > 0$. From the first three equations $x^2 = y^2 = z^2$, and since x, y, and z are positive, $x = y = z$. Then, from the fourth equation, $x = y = z = 1/3$ and $F(1/3, 1/3, 1/3) = 1/9$ is a constrained extremum. Since $(1/2, 1/4, 1/4)$ satisfies the constraint and $F(1/2, 1/4, 1/4) = 5/32 > 1/9$, $F(1/3, 1/3, 1/3) = 1/9$ is a constrained minimum.

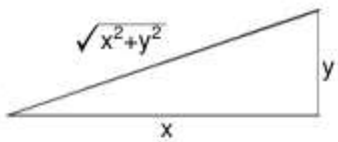
18. $F_x = 8xy^2z^2$; $F_y = 8x^2yz^2$; $F_z = 8x^2Y^2z$; $g_x = 2x$; $g_y = 2y$; $g_z = 2z$. We need to solve $8xy^2z^2 = 2\lambda x$, $8x^2yz^2 = 2\lambda y$, $8x^2y^2z = 2\lambda z$ or $4x^2y^2z^2 = \lambda x^2$, $4x^2y^2z^2 = \lambda y^2$, $4x^2y^2z^2 = \lambda z^2$ along with $x^2 + y^2 + z^2 - 9 = 0$ or $x^2 + y^2 + z^2 = 9$ for $x > 0$, $y > 0$, $z > 0$, and hence $lambda > 0$. From the first three equations, we see $x^2 = y^2 = z^2$. Substituting into the third equation, we obtain $x^2 + x^2 + x^2 = 3x^2 = 9$. Thus, since x, y, and z are positive, $x = y = z = \sqrt{3}$ and $F(\sqrt{3}, \sqrt{3}, \sqrt{3}) = 108$ is a constrained extremum. Since $(1, 2, 2)$ satisfies the constraint and $F(1, 2, 2) = 64 < 108$, $F(\sqrt{3}, \sqrt{3}, \sqrt{3}) = 108$ is a constrained maximum.

19. $F_x = 2x$; $F_y = 2y$; $F_z = 2z$; $g_x = 2$; $g_y = 1$; $g_z = 1$; $h_x = -1$; $h_y = 2$; $h_z = -3$. We need to solve $2x = 2\lambda - \mu$, $2y = \lambda + 2\mu$, $2z = \lambda - 3\mu$ subject to $2x + y + z = 1$, $-x + 2y - 3z = 4$. Solving the first three equations for x, y, and z, respectively, and substituting into the constraint equations, we obtain $2\lambda - \mu + \lambda/2 - 3\mu/2 = 1$, $-\lambda + \mu/2 + \lambda + 2\mu - 3\lambda/2 + 9\mu/2 = 4$ or $6\lambda - 3\mu = 2$, $-3\lambda + 14\mu = 8$. From this, we obtain $\lambda = 52/75$ and $\mu = 54/75$. Then $x = 1/3$, $y = 16/15$, and $z = -11/15$. Thus, $F(1/3, 16/15, -11/15) = 134/75$ is a constrained minimum.

20. $F_x = 2x$; $F_y = 2y$; $F_z = 2z$; $g_x = 4$; $g_y = 0$; $g_z = 1$; $h_x = 2x$; $h_y = 2y$; $h_z = -2z$. We need to solve $2x = 4\lambda + 2x\mu$, $2y = 2y\mu$, $2z = \lambda - 2z\mu$ subject to $4x + z = 7$, $z^2 = x^2 + y^2$.

Consider the second equation. If $y = 0$, then the constraint equations become $4x + z = 7$ and $z^2 = x^2$. The solutions of these equations are $x = z = 7/5$ and $x = -z = 7/3$. In either case, the first and third equations can be solved for λ and μ. Thus, $(7/5, 0, 7/5)$ and $(7/3, 0, -7/3)$ are candidates for constrained extrema. Now, if $y \neq 0$, then from the second equations $\mu = 0$. In this case, $2x = 4\lambda$ and $2z = \lambda$ or $x = 4z$. Then the first constraint equation becomes $16z + z = 17z = 7$, so $z = 7/17$. Then, $x = 28/17$ and $y^2 = z^2 - x^2 < 0$. Hence, the system has no solution when $y \neq 0$. Thus, $F(7/5, 0, 7/5) = 98/25$ is a constrained minimum and $F(7/3, 0, -7/3) = 98/9$ is a constrained maximum.

21. We want to maximize $A(x, y_x y/2$ subject to $P(x, y) = x + y + \sqrt{x^2 + y^2} - 4 = 0$. $A_x = y/2$; $A_y = x/2$; $P_x = 1 + x/\sqrt{x^2 + y^2}$; $P_y = 1 + y/\sqrt{x^2 + y^2}$. We need to solve $y/2 = \lambda + \lambda x/\sqrt{x^2 + y^2}$, $x/2 = \lambda + \lambda y/\sqrt{x^2 + y^2}$, $x + y + \sqrt{x^2 + y^2} - 4 = 0$ for $x > 0$, $y > 0$, and hence $\lambda > 0$. Subtracting the second equation from the first, we have $(y - x)/2 = \lambda(x - y)/\sqrt{x^2 + y^2}$ or $(y - x) = (y - x)(-2\lambda/\sqrt{x^2 + y^2})$. Since $-2\lambda/\sqrt{x^2 + y^2}$ is negative, it cannot equal 1, and hence $y - x = 0$ or $y = x$. Substituting in the third equation gives $2x + \sqrt{2x^2} = (2 + \sqrt{2})x = 4$. Thus, $x = y = 4/(2 + \sqrt{2})$ and this maximum area is $A(4/(2 + \sqrt{2}), 4/(2 + \sqrt{2})) = 4/(3 + 2\sqrt{2})$.



22. Let the base of the box have dimensions x and y and let the height be z. We want to maximize $V(x, y, z) = xyz$ subject to $S(x, y, z) = xy + 2yz + 2xz - 75 = 0$. Now $V_x = yz$; $V_y = xz$; $V_z = xy$; $S_x = y + 2z$; $S_y = x + 2z$; $S_z = 2y + 2x$. We need to solve $yz = \lambda(y + 2z)$, $xz = \lambda(x + 2z)$, $xy = \lambda(2y + 2x)$, $xy + 2yz + 2xz - 75 = 0$ or $xyz = \lambda(xy + 2xz)$, $xyz = \lambda(xy + 2yz)$, $xyz = \lambda(2yz + 2xz)$, $xy + 2yz + 2xz = 75$, for $x > 0$, $y > 0$, $z > 0$, and thus $\lambda > 0$. From the first three equations, we have $xy + 2xz = xy + 2yz$, which gives $xz = yz$ or $x = y$; and $xy + 2yz = 2yz + 2xz$, which gives $xy = 2xz$ or $y = 2z$. Substituting $x = y = 2z$ into the fourth equation, we obtain $4z^2 + 4z^2 + 4z^2 = 12z^2 = 75$. Thus, $z = 5/2cm$ and $x = y = 5cm$. When the box is closed, $S(x, y, z) = 2(xy + yz + xz) - 75$, $S_x = 2(y + z)$, $S_y = 2(x + z)$, $S_z = 2(x + y)$, and we need to solve $xyz = 2\lambda(xy + xz)$, $xyz = 2\lambda(xy + yz)$, $xyz = 2\lambda(yz + xz)$, $2(xy + yz + xz) = 75$ for $x > 0$, $y > 0$, $z > 0$, and thus $\lambda > 0$. From the first three equations, we have $xy + xz = xy + yz$, which gives $xz = yz$ or $x = y$; and $xy + yz = yz + xz$, which gives $xy = xz$ or $y = z$. Substituting $x = y = z$ into the fourth equation, we obtain $2(x^2 + x^2 + x^2) = 6x^2 = 75$. Thus, $x = y = z = 5/\sqrt{2}$. The box is a cube with each side $5/\sqrt{2}cm$.

23. We want to maximize $V(x, y) = 9\pi x + 3\pi y$ subject to $S(x, y) = 9\pi + 6\pi x + 3\pi\sqrt{9 + y^2} - 81\pi$. Now $V_x = 9\pi$; $V_y = 3\pi$; $S_x = 6\pi$; $S_y = 3\pi y/\sqrt{9 + y^2}$. We need to solve $9\pi = 6\pi\lambda$, $3\pi = 3\pi\lambda y/\sqrt{9 + y^2}$, $9\pi + 6\pi x + 3\sqrt{9 + y^2} - 81\pi = 0$ for $x > 0$ and $y > 0$. From the first equation, $\lambda = 3/2$. Using $\lambda = 3/2$ in the second equation gives $y = 6/\sqrt{5}$. From the third equation, we have $6x + 3\sqrt{9 + 36/5} = 72$ or $x = 12 - 9/2\sqrt{5}$. The volume is maximum when $x = 12 - 9/2\sqrt{5}m$ and $y = 6/\sqrt{5}m$.

24. $U_x = \dfrac{1}{3}x^{-2/3}y^{2/3}$; $U_y = \dfrac{2}{3}x^{1/3}y^{-1/3}$; $g_x = 1$; $g_y = 6$. We need to solve $\dfrac{1}{3}x^{-2/3}y^{2/3} = \lambda$, $\dfrac{2}{3}x^{1/3}y{-}1/3 = 6\lambda$, $x + 6y - 18 = 0$ or $y = 3\lambda x^{2/3}y^{1/3}$, $\dfrac{1}{3}x = 3\lambda x^{2/3}y^{1/3}$, $x + 6y = 18$. From the first two equations, $y = x/3$. Substituting into the third equation, we

have $x + 2x = 3x = 18$. Thus, $x = 6$ and $y = 2$. Since $(12, 1)$ satisfies the constraint and $U(12, 1) = 12^{1/3}$, $U(6, 2) = 6^{1/3}2^{2/3} = 24^{1/3}$ is a constrained maximum.

25. We want to maximize $z(x, y) = P - x - y$ subject to $z^2/xy^3 = k$ or $(P - x - y)^2 - kxy^3 = 0$. Now $z_x = -1$; $z_y = -1$; $g_x = -2(P - x - y) - ky^3$; $g_y = -2(P - x - y) - 3kxy^2$. We need to solve $-1 = -2\lambda(P - x - y) - \lambda ky^3$, $-1 = -2\lambda(P - x - y) - 3\lambda kxy^2$, $(P - x - y)^2 = kxy^3$ for $x > 0$, $y > 0$, and $z > 0$. From the first two equations, we have $y = 3x$. Substituting into the third equation, we obtain $(P - 4x)^2 = 27kx^4$ or $\sqrt{27k}x^2 = P - 4x$. (Since $z > 0$, $z = P - x - y = P - 4x > 0$.) Using the quadratic formula and the fact that $x > 0$, we find

$$x = \frac{-4 + \sqrt{16 + 4P\sqrt{27k}}}{2\sqrt{27k}} = \frac{-2 + \sqrt{4 + P\sqrt{27k}}}{27k}.$$

Then the maximum value of z is $P - 4x = P + 4(2 - \sqrt{4 + P\sqrt{27k}}/\sqrt{27k}$.

26. (a) See part **(b)**.

    (b) Maximizing $1/(x_1^2 \cdots x_n^2)$ is equivalent to minimizing the denominator $F(x_1, \ldots x_n) = x_1^2 + \cdots + x_n^2$. The constraint is still $x_1 + \cdots x_n = 1$, which we can write as $g(x_1, \ldots x_n) = x_1 + \cdots + x_n - 1 = 0$. Since $\partial F/\partial x_i = 2x_i$ and $\partial g/\partial x_i = 1$, we can get the equations $2x_1 = \lambda,$, $2x_2 = \lambda, \ldots, 2x_n = \lambda$, $x_1 + \cdots x_n = 1$. The solution is $x_1 = \ldots = 1/2$ (and $\lambda = 1/2n$).

27. $f(x, y)$ is the square of the distance from a point on the graph of $x^4 + y^4 = 1$ to the origin. The points $(0, \pm 1)$ and $(\pm 1, 0)$ are closest to the origin, while $(\pm 1/\sqrt[4]{2}, \pm 1/\sqrt[4]{2})$ are farthest from the origin.

28. $F(x, y, z)$ is the square of the distance from a point on the plane $x + 2y + 3z = 4$ to the origin. The point $(2/7, 4/7, 6/7)$ is closest to the origin.

29. F is the square of the distance of points on the intersection of the planes $2x + y + z = 1$ and $-x + 2y - 3z = 4$ from the origin. The point $(1/3, 16/15, -11/15)$ is closest to the origin.

30. F is the square of the distance of points on the intersection of the plane $4x + z = 7$ and the circular cone $z^2 = x^2 = y^2$. The point $(7/5, 0, 7/5)$ is closest to the origin and the point $(7/3, 0, -7/3)$ is farthest from the origin.

31. We want to minimize $f(x,) = x^2 + y^2$ subject to $xy^2 = 1$. Now $f_x = 2x$; $f_y = 2y$; $g_x = y^2$; $g_y = 2xy$. We need to solve $2x = \lambda y^2$, $2y = 2\lambda xy$, $xy^2 - 1 = 0$ or $2xy = \lambda y^3$, $2xy = 2\lambda x^2 y$, $xy^2 = 1$ for $x > 0, y > 0$, and hence $\lambda > 0$. From the first two equations, we have $y^3 = 2x^2y$ or $y^2 = 2x^2$. Substituting into the third equation gives $2x^3 = 1$ or $x = 2^{-1/3}$. Again, from the third equation we have $y = 1/(2^{-1/3})^{1/2} = 2^{1/6}$. Thus, the point closest to the origin is $(2^{-1/3}, 2^{1/6})$. Since the surface is $F(x, y, z) = xy^2 - 1 = 0$, $\nabla F = y^2\mathbf{i} + 2xy\mathbf{j}$ is normal to the surface at (x, y, z). Thus, a normal to the surface at $(2^{-1/3}, 2^{1/6}, 0)$ is $\nabla F(2^{-1/3}, 2^{1/6}, 0)$ is $\nabla F(2^{-1/3},\ 21/6, 0) = 2^{1/3}\mathbf{i} + 2(2^{-1/3})(2^{1/6})\mathbf{j} = 2^{1/3}\mathbf{i} + 2^{5/6}\mathbf{j} = 2^{2/3}(2^{-1/3}\mathbf{i} + 2^{1/6}\mathbf{j})$. Since $\nabla F(2^{-1/3}, 2^{1/6}, 0)$ is a multiple of the vector from the origin to $P(2^{-1/3}, 2^{1/6}, 0)$, this vector is perpendicular to the surface.

32. $F_x = \frac{1}{3}x^{-2/3}y^{1/3}z^{1/3}$; $F_y = \frac{1}{3}x^{1/3}y^{-2/3}z^{1/3}$; $F_z = \frac{1}{3}x^{1/3}y^{1/3}z^{-2/3}$; $g_x = 1$; $g_y = 1$; $g_z = 1$. We need to solve $\frac{1}{3}x^{-2/3}y^{1/3}z^{1/3} = \lambda$, $\frac{1}{3}x^{1/3}y^{-2/3}z^{1/3} = \lambda$, $\frac{1}{3}x^{1/3}y^{1/3}z^{-2/3} = \lambda$, $x + y + z - k = 0$ or $x^{1/3}y^{1/3}Z1/3 = 3\lambda x$, $x^{1/3}y^{1/3}z^{1/3} = 3\lambda y$, $x^{1/3}y^{1/3}z^{1/3} = \lambda z$, $x + y + z = k$. From the first three equations, $x = y = z$. Substituting into the fourth equation, $x + x + x = 3x = k$. Thus, $x = y = z = k/3$ and $F(k/3, k/3, k/3) = k/3$ is a constrained maximum.

33. For any $x + y + z = k$, by Problem 32, $\sqrt[3]{xyz} \leq \frac{k}{3} = \frac{x+y+z}{3}$.

34. Distance from the xz-plane is measured by $|y|$. Alternatively, we will find the extreme values of $F(x, y, z) = y^2$ subject to $g(x, y, z) = x^2 + z^2 - 1 = 0$ and $h(x, y, z) = x + y + 2z - 4 = 0$. Now $F_x = 0$; $F_y = 2y$; $F_z = 0$; $g_x = 2x$; $g_y = 0$; $g_z = 2z$; $h_x = 1$; $h_y = 1$; $h_z = 2$. We need to solve $0 = 2\lambda x + \mu$, $2y = \mu$, $0 = 2\lambda z + 2\mu$, $x^2 + z^2 = 1$, $x + y + 2z = 4$. By inspection we see that if $\lambda = 0$, then $\mu = 0$ and $y = 0$. Similarly, if $\mu = 0$, then $\lambda = 0$ and $y = 0$. Substituting $y = 0$ into the fourth and fifth equations, we obtain the system $x^2 + z^2 = 1$, $x + 2z = 4$, which is inconsistent. Thus, $\mu \neq 0$ and $\lambda \neq 0$. Now, solving the first three equations for x, y, and z and substituting into the fourth and fifth equations, we obtain the system

$$\frac{\mu^2}{4\lambda^2} + \frac{\mu^2}{\lambda^2} = 1, \quad -\frac{\mu}{2\lambda} + \frac{\mu}{2} - \frac{2\mu}{\lambda} = 4 \text{ or } 5\mu^2 = 4\lambda^2, \quad (\lambda - 5)\mu = 8\lambda.$$

Solving the second equation for μ and substituting into the first, we obtain $5(\frac{8\lambda}{\lambda - 5})^2 = 4\lambda^2$ or $80 = (\lambda - 5)^2$. Thus, $\lambda = 5 \pm 4\sqrt{5}$. From $\mu = 8\lambda/(\lambda - 5)$ we find that corresponding values of μ are $8 \pm 2\sqrt{5}$. Since $2y = \mu$ we see that the objective function $F(x, y, z) = y^2$ is minimized when $\mu = 8 - 2\sqrt{5}$ and maximized when $\mu = 8 + 2\sqrt{5}$. Corresponding values of x, y, and z are $x = -\frac{\mu}{2\lambda} = -\frac{8 \pm 2\sqrt{5}}{10 \pm 8\sqrt{5}} = \mp\frac{1}{\sqrt{5}} \approx |mp0.45$, $y - \mu/2 = 4 \pm \sqrt{5} \approx 6.24$ and 1.76, $z = -\mu/\lambda = 2x = \mp 2/\sqrt{5} \approx \mp 0.89$. The closest point is $(-1/\sqrt{5}, 4 - \sqrt{5}, -2\sqrt{5})$ or about $(-4.5, 6.24, -0.89)$.